

```
In [ ]: # Cell 1 – Imports  
import subprocess, sys, os  
from pathlib import Path  
import marimo as mo
```

```
In [ ]: # # Cell 3 #training dont run  
# from ultralytics import YOLO  
  
# model = YOLO("yolov8n.pt") #wts dwld  
# results = model.train(  
#     data="VOC.yaml",  
#     epochs=20,  
#     imgsz=640,  
#     batch=16,  
#     device=0,  
#     project="runs/detect",  
#     name="voc_train"  
# )
```

```
In [ ]: # Cell 4 – Evaluate best checkpoint  
from ultralytics import YOLO  
  
model = YOLO("/home/aswin/runs/detect/runs/detect/voc_train2/weights/best.pt")  
  
metrics = model.val(data="VOC.yaml", device=0)  
print(f"Precision:      {metrics.box.mp:.3f}")  
print(f"Recall:         {metrics.box.mr:.3f}")  
print(f"mAP@0.5:         {metrics.box.map50:.3f}")  
print(f"mAP@0.5:0.95:    {metrics.box.map:.3f}")
```

Ultralytics 8.4.33 🚀 Python-3.11.14 torch-2.10.0+cu128 CUDA:0 (NVIDIA GeForce RTX 5050 Laptop GPU, 7708MiB)

Model summary (fused): 73 layers, 3,009,548 parameters, 0 gradients, 8.1 GFLOPs

val: Fast image access ✅ (ping: 0.0±0.0 ms, read: 3807.1±632.9 MB/s, size: 75.9 KB)

val: Scanning /home/aswin/datasets/VOC/labels/test2007.cache... 4952 images, 0 backgrounds, 0 corrupt: 100% ————— 4952/4952 1.5Git/s 0.0s

	Class	Images	Instances	Box(P	R	mAP50
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mAP50-95): 1%	Class	2/310	4.6it/s	0.2s<1:08	R	mAP50
mAP50-95): 1%	Class	4/310	6.2it/s	0.4s<49.5s	R	mAP50
mAP50-95): 2%	Class	7/310	11.4it/s	0.6s<26.6s	R	mAP50
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		271/310	12.6it/s	18.2s<3.1s		

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	Class	Images	Instances	Box(P	R	mAP50
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mAP50-95): 98%	Class	Images	Instances	Box(P	R	mAP50
mAP50-95): 99%	Class	Images	Instances	Box(P	R	mAP50
mAP50-95): 100%	Class	Images	Instances	Box(P	R	mAP50
mAP50-95): 100%	Class	Images	Instances	Box(P	R	mAP50
	all	4952	12032	0.823	0.748	0.835
0.629	aeroplane	204	285	0.912	0.8	0.887
0.652	bicycle	239	337	0.945	0.81	0.909
0.689	bird	282	459	0.807	0.683	0.802
0.56	boat	172	263	0.766	0.639	0.743
0.497	bottle	212	469	0.778	0.635	0.74
0.515	bus	174	213	0.871	0.836	0.903
0.778	car	721	1201	0.886	0.857	0.931
0.743						

0.716	cat	322	358	0.899	0.817	0.892
0.469	chair	417	756	0.704	0.583	0.669
0.652	cow	127	244	0.785	0.779	0.856
0.62	diningtable	190	206	0.776	0.725	0.811
0.686	dog	418	489	0.85	0.755	0.875
0.735	horse	274	348	0.889	0.862	0.934
0.662	motorbike	222	325	0.889	0.785	0.899
0.647	person	2007	4528	0.896	0.785	0.903
0.35	pottedplant	224	480	0.69	0.51	0.587
0.65	sheep	97	242	0.721	0.81	0.841
0.664	sofa	223	239	0.689	0.724	0.801
0.66	train	259	282	0.865	0.855	0.894
0.64	tvmonitor	229	308	0.84	0.718	0.825

Speed: 0.2ms preprocess, 2.5ms inference, 0.0ms loss, 0.4ms postprocess per image

Results saved to **/home/aswin/runs/detect/val5**

Precision: 0.823

Recall: 0.748

mAP@0.5: 0.835

mAP@0.5:0.95: 0.629

The settings for which I ran this for. epochs=20 imgsz=640 batch=16

- I got precision of 0.823 which is really good
- Recall of 0.748, meaning some missed detection is there.
- mAP@0.5 got 0.835 meaning high detection capability.
- mAP@0.5:0.95 got 0.629 meaning moderate localization precision
- So these says our bounding boxes are coarse and the precision is not that good. This can be due to the low number of epochs as well as the smaller yolo model being used to train without much computational expense.
- Strong classes obtained are - car(0.93), horse(0.93), bus(0.93) and person(0.90). This is because higher representation in our VOC dataset, relatively large objects and easy distinguishability.